Parametrized Lines and Intersection

Exercise

$$L_1 : \begin{cases} x = 2 - t \\ y = 1 + t \end{cases} \qquad L_2 : \begin{cases} x = 2 + u \\ y = 4 + 2u \end{cases}$$

Do they intersect? If so, find the point.

To find the intersection, equate coordinates:

$$2 - t = 2 + u$$

$$1 + t = 4 + 2u$$

From the first equation:

$$-t = u \Rightarrow u = -t$$

Substitute into the second equation:

$$1 + t = 4 + 2(-t)$$

$$1 + t = 4 - 2t$$

$$3t = 3 \Rightarrow t = 1$$

Then:

$$u = -1$$

Substitute into one of the lines:

$$x = 2 - 1 = 1, \quad y = 1 + 1 = 2$$

Intersection point:

$$(1, 2)$$

Exercise 1

Write parametric equations for a line through the point

$$P_0 = (1, 2, 3)$$

and parallel to the vector

$$\mathbf{v} = (1, 3, 5).$$

Solution

Let $P = (x, y, z)$ be a point on the line. Then the vector

$$\overrightarrow{P_0P} = (x - 1,\ y - 2,\ z - 3)$$

is parallel to $(1, 3, 5)$.

So we write:

$$(x - 1,\ y - 2,\ z - 3) = t(1, 3, 5)$$

This gives:

$$\begin{cases} x - 1 = t \\ y - 2 = 3t \\ z - 3 = 5t \end{cases}$$

Hence, the parametric equations are:

$$\begin{cases} x = 1 + t \\ y = 2 + 3t \\ z = 3 + 5t \end{cases}$$

Example 2

If the direction vector is

$$(2, 6, 10) = 2\mathbf{v},$$

we obtain the same line with a different parametrization.

That is, the moving point follows the same path as in Exercise 1, but reaches each point at a different time.

Example 3

In general, the line through

$$P_0 = (x_0, y_0, z_0)$$

in the direction of (i.e., parallel to)

$$\mathbf{v} = (v_1, v_2, v_3)$$

has parametrization:

$$(x, y, z) = (x_0 + tv_1,\ y_0 + tv_2,\ z_0 + tv_3)$$

Equivalently,

$$\begin{cases} x = x_0 + tv_1 \\ y = y_0 + tv_2 \\ z = z_0 + tv_3 \end{cases}$$

Example 4

Find the line through the points

$$P_0 = (1, 2, 3) \quad \text{and} \quad P_1 = (2, 5, 8).$$

Solution:

We use the given data to find the basic elements (a point and a direction vector).

A point is:

$$P_0 = (1, 2, 3)$$

The direction vector is:

$$\mathbf{v} = \overrightarrow{P_0P_1} = (2 - 1,\ 5 - 2,\ 8 - 3) = (1, 3, 5)$$

Thus, the parametric equation of the line is:

$$(x, y, z) = (1, 2, 3) + t(1, 3, 5)$$

$$= (1 + t,\ 2 + 3t,\ 3 + 5t)$$

Hence,

$$\begin{cases} x = 1 + t \\ y = 2 + 3t \\ z = 3 + 5t \end{cases}$$

Parametric Equations of Lines

1) Give parametric equations for x, y, z on the line through $(1, 2, 3)$ in a direction parallel to $\langle 2, -3, -1 \rangle$.

2) Give parametric equations for the intersection of the planes

$$x + y + z = 1$$

and

$$x + 2y + 3z = 2.$$

Solution

1) We are given a point and a direction vector.

$$(x, y, z) = (1, 2, 3) + t(2, -3, -1)$$

So,

$$x = 1 + 2t, \quad y = 2 - 3t, \quad z = 3 - t$$

2) To find the line of intersection:
Let $z = 0$:

$$x + y = 1$$

$$x + 2y = 2$$

Solving:

$$y = 1, \quad x = 0$$

So a point is:

$$(0, 1, 0)$$

Direction vector = cross product:

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix} = (1, -2, 1)$$

Parametric equations:

$$x = t, \quad y = 1 - 2t, \quad z = t$$

Intersection of a Line and a Plane

Question: Where does the line through $(0, -1, 1)$ and $(2, 3, 3)$ meet the plane with equation

$$2x + y - z = 1\,?$$

Solution

Let

$$P_1 = (0, -1, 1), \quad P_2 = (2, 3, 3)$$

Line equation:

$$P_1 + t(P_2 - P_1)$$

$$= (0, -1, 1) + t(2, 4, 2)$$

Parametric form:

$$(x, y, z) = (2t,\ 4t - 1,\ 2t + 1)$$

Substitute into plane equation:

$$2x + y - z = 1$$

$$2(2t) + (4t - 1) - (2t + 1) = 1$$

$$4t + 4t - 1 - 2t - 1 = 1$$

$$6t - 2 = 1$$

$$6t = 3 \Rightarrow t = \frac{1}{2}$$

Point of intersection:

$$(2t,\ 4t - 1,\ 2t + 1) = (1, 1, 2)$$

Another Example

Consider the plane:

$$P:\ 2x + y - 4z = 4$$

Find the point of intersection of P with the line:

$$x = t, \quad y = 2 + 3t, \quad z = t$$

Continuation

(b) Find all points of intersection of the line

$$x = 1 + t, \quad y = 4 + 2t, \quad z = t$$

with the plane

$$2x + y - 4z = 4.$$

Solution:
Substitute into the plane equation:

$$2(1 + t) + (4 + 2t) - 4(t) = 4$$

$$2 + 2t + 4 + 2t - 4t = 4$$

$$6 = 4$$

This is false, so there are **no values of** t.

$$\Rightarrow \text{No points of intersection.}$$

(c) Find all points of intersection of the line

$$x = t, \quad y = 4 + 2t, \quad z = t$$

with the plane
$$2x + y - 4z = 4.$$

Solution:
Substitute into the plane equation:

$$2(t) + (4 + 2t) - 4(t) = 4$$

$$2t + 4 + 2t - 4t = 4$$

$$4 = 4$$

This is true for **all values of** t.

$$\Rightarrow \text{The line lies entirely in the plane.}$$

So, **all points of the line are points of intersection**.